RAKESH BRAHMA

Agentic AI Engineer | RAG & LLM Systems | B.Tech Computer Science + MBA (IIM Indore)

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Computer Science engineer (B.Tech, MBA IIM Indore) who spent 8 years running commercial operations at IndianOil, a regulated PSU: channel-partner management, B2C and B2B sales to fleets and institutions, government liaison, multi-crore accounts. Now builds agentic AI and RAG systems for the same kind of rule-bound, document- and number-heavy work. IIT Roorkee-certified in Agentic AI; ships systems with evidence attached: golden-set evaluation, ablations, and deterministic guards so the LLM never produces a number on its own.

AI PROJECTS

Lending Support Agent | Lending-operations assistant (CrewAI + AutoGen) | *GitHub* **2026**

- Lending-operations desk assistant: a CrewAI triage step routes each question to loan lookup, policy RAG over 12 documents (ChromaDB, BGE embeddings) or an EMI calculator; an AutoGen reviewer team checks every reply. FastAPI REST + WebSocket.
- Guardrails: PAN, Aadhaar and account numbers masked; prompt injection refused; numbers only from tool output; read-only (deletes blocked, changes routed to a person). Runs on a mock LLM with zero API keys; 627 tests passing.

Advanced Multimodal RAG | Annual-report analysis | *GitHub* **2026**

- Q&A over a 175-page annual report (NVIDIA FY2026 10-K) indexing text, financial tables and charts in one ChromaDB index: Docling parsing cross-checked by pdfplumber, vision-LLM figure descriptions, page-level citations, FastAPI backend.
- Justified each component by measurement: hybrid BM25 + dense retrieval (RRF) and glossary query expansion lifted recall@8 from 0.52 to 0.95 at zero model calls; a reranker that lowered recall was left off. 35 of 35 numeric answers exact with Claude Sonnet, about 94% with the open model on Groq; RAGAS faithfulness 0.99 on a 48-question golden set.
- Two-tier semantic cache with an 8-key slot guard: 0 false hits on 286 adversarial pairs while catching 56% of paraphrases, served in 35 ms vs 2.2 s cold (Groq); deterministic formula engine and a number verifier; 97% test coverage on the deterministic core.

Fleet Profitability Agent | Fuel & toll route-cost optimizer | *GitHub* **2026**

- LangGraph agent that prices every viable route for a long-haul Indian freight trip: truck-aware routing, district diesel prices, official NETC toll data and a refuel optimizer (where and how much to fill); CLI, FastAPI and Streamlit demo.
- LLM kept off the money path: no figure in the plan comes from the language model. Evaluated on 5 real corridors x 3 tank sizes: 3 of 5 clear the INR 1/L savings target, zero-saving cases reported, not hidden. Built on my fuel-retail work at IndianOil.

AML Compliance Agent | Sanctions-screening agent | *GitHub* **2026**

- RAG + tool-calling agent screening cross-border payments against a written AML policy (ChromaDB) and the live OpenSanctions API, returning ALLOW / REVIEW / BLOCK verdicts with an evidence trail (FastAPI, Docker, Groq/Llama 3.3). Risk-based screening on the OpenSanctions topics field stops false blocks on PEP records; threshold arithmetic kept in code.

Port & Customs Clearance Copilot | Multi-agent shipment clearance | *GitHub* **2026**

- Five agents, each with its own MCP toolbox, check a shipment's papers as soon as they exist: invoice, packing list and bill of lading cross-checked, demurrage priced and the customs rulebook queried by RAG in parallel.
- Two human gates (every filing, every rule change) with Postgres checkpoints; a number guard blocks LLM figures not found in the data. INR 82,212 demurrage avoided in a four-container demo; rulebook hit@1 of 1.0 (12-question golden set); 619 tests.

Additional builds: PolicyMind (live demo; hybrid BGE + BM25 RAG, reranking and parent-child chunking over a 183-page PSU policy); Rivaya Returns (n8n + FastAPI returns adjudication, human-in-the-loop); Tender Q&A bot (FAISS + BM25 RAG with abstention and citation eval).

PROFESSIONAL EXPERIENCE

Manager, Retail Sales | IndianOil Corporation Ltd. | *Arunachal Pradesh, Bhutan, NE Region* **Sep 2018 - Present**

- Drove HSD volume growth from 2% to 10% YoY (an 8-point jump, about 3,672 KL gain) to win highest customer acquisition and volume growth in the FY25-26 Xtrapower fleet program, IOCL's flagship.
- Own an INR 135 crore average monthly turnover across petrol and diesel, leading 64 channel partners over a 9-district territory including 3 sensitive Indo-China border districts.
- Closed the region's largest single B2B consignment to an international Oil Marketing Company: 26 KL of Servo lubricants into Bhutan, a first-time market entry, through Ministry liaison and outlet upgradation.
- Built an Excel sales-analysis model for YoY throughput and gap analysis; piloted CNG commissioning in Lower Assam with Adani Gas.
- Recognized at the Annual Field Officers' Conclave (FY22-23) for NE-Region EV-charger commissioning and flagship-outlet upgrades.

Assistant Manager, Business Development | The Q Experiences | *Mumbai* **Mar 2017 - Jan 2018**

- Member of the founding team that chartered India's first luxury cruise to Antarctica (in alliance with Ponant); ran website/online presence, B2B corporate communications and a LinkedIn lead-generation campaign. Venture later won a World Travel Award.

EDUCATION & CERTIFICATION

Certification in Agentic AI Systems & Design | iHUB DivyaSampark, IIT Roorkee (via Masai School) **2026**

MBA, General Management (Marketing & Operations) | Indian Institute of Management, Indore **2015 - 2017**

B.Tech, Computer Science Engineering | Guru Gobind Singh Indraprastha University, Delhi **2011 - 2015**

TECHNICAL SKILLS

AI / LLM: Agentic AI, Multi-agent Orchestration, LangGraph, LangChain, CrewAI, AutoGen, MCP (Model Context Protocol), Tool / Function Calling, RAG (hybrid retrieval, BM25, RRF, query expansion, reranking, chunking), Multimodal RAG (vision LLMs, tables & charts), Semantic Caching, Structured Outputs (Pydantic), Prompt Engineering, Human-in-the-Loop

Evaluation & Reliability: RAG Evaluation (RAGAS, golden datasets, recall@k, MRR), Ablation Studies, Output Verification & Guardrails, PII Masking, pytest, Test Coverage Gates

Engineering: Python, FastAPI, REST APIs, WebSocket, Docker, Git / GitHub, Streamlit, n8n, ChromaDB, FAISS, SQL (PostgreSQL), Docling / pdfplumber, Groq & Anthropic Claude APIs, Render, Streamlit Cloud

Business & Data: P&L Ownership, Channel & Territory Management, Stakeholder & Government Liaison, Go-to-Market, B2B Sales, Cross-functional Leadership, Excel modelling & linear programming, Power BI, SAP (SD), ZOHO CRM